DAISY ZHOU

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EDUCATION

Columbia University — New York, NY
MS in Operations Research, GPA 3.75 — Aug 2025 - Dec 2026
- Coursework: Dynamic Pricing & Revenue Management, Transportation Analytics & Logistics, Statistical Learning & Machine Learning, Applied Data Science, Stochastic Models, Optimization Programming (Linear, Integer, Nonlinear, Network)

Georgia Institute of Technology — Atlanta, GA
BS in Industrial Engineering, GPA 4.0 — Dec 2022

WORK EXPERIENCE

The Boeing Company — Everett, WA
Systems Engineer — Feb 2023 - Jun 2025
- Modeled SysML functional and logical architecture for fuselage assembly and 737 wing panel assembly manufacturing lines in Cameo to consolidate traditional document-based materials into a digitalized single source of truth
- Co-authored 4 user guides on building logical and functional architecture, state machines, and manufacturing bill of materials in Cameo to drive data consistency and uniform standards across all Production System use cases
- Drove pathfinding efforts with domain experts to determine best modeling practices on Production System design and development, advancing Model Based Systems Engineering maturity and department-wide adoption
- Developed and implemented custom data queries in Cameo to build standardized table templates to optimize data reporting processes and replace error-prone Excel artifacts

Delta Air Lines — Atlanta, GA
Reservations Technology and Data Analytics Intern — May 2021 - Aug 2021
- Performed data cleaning and generated Excel visualizations on annual work schedule data for 20+ employees
- Developed reporting tool using Excel VBA to display current resource capacity levels, reducing report generation time by 95%
- Analyzed resource supply and demand imbalance, flagged work capacity exceeding 80% utilization, and collaborated with senior analysts to deliver data-driven insights for leadership’s long-term resource planning

ACADEMIC & PROJECT EXPERIENCE

Course Assistant: Optimization Models and Methods — Jan 2026 - May 2026
- Graded quizzes and assignments, proctored exams, and assisted students with questions on linear, mixed integer, nonlinear, and network programming concepts

Comparative Survival Modeling for Incident Myocardial Infarction (MI) Prediction — Feb 2026 - May 2026
- Built Cox PH, Elastic Net Cox, Random Survival Forest, and Gradient Boosting survival models to to predict time-to-MI among 33k+ ICU patients using MIMIC-IV clinical data, achieving C-index values near 0.70
- Engineered high-dimensional clinical features from ICU vitals, labs, comorbidities, and medications collected during the first 24 hours of ICU admission

IBM Job Salary Prediction and Labor Market Analytics — April 2026 - May 2026
- Developed end-to-end ML pipeline on large-scale web-scraped IBM job posting data using feature engineering, clustering, and regression modeling to analyze salary and labor market trends
- Built and evaluated Gradient Boosting, Random Forest, and Ridge Regression models using cross-validation and hyper parameter tuning, achieving R^2 = 0.85 for salary prediction

Dynamic Pricing and Revenue Optimization Course Project — April 2026 - May 2026
- Developed multi-phase stochastic pricing framework integrating Bayesian demand learning, Thompson Sampling, Histogram Gradient Boosting, and kernel-based local demand estimation for revenue optimization under nonstationary market conditions
- Formulated competitor-aware pricing polices using contextual feature engineering, Euclidean distance—based market state matching, and adaptive expected-revenue maximization across a dynamically evolved competitive environment simulation

Columbia Transportation Logistics Project — Nov 2025 - Dec 2025
- Designed dynamic batching and routing optimization model for on-demand micro-transit shuttle using pickup-and-delivery TSP on real road networks, reducing average passenger wait time by 43% without increasing fleet size
- Performed model validation and sensitivity analyses to evaluate system-level performance tradeoffs through end-to-end simulation using historical ride request data and OpenStreetMap network data

New York State Child Care Desert Elimination Project — Sep 2025 - Oct 2025
- Developed mixed-integer programming model to eliminate all child care deserts across 1300+ NY zip codes while optimizing facility construction and capacity expansion under budget, regulatory, and spatial constraints

Teaching Assistant: Constraint Programming — Aug 2022 - Dec 2022
- Co-led twice-weekly interactive sessions for 200+ undergraduate and graduate students, graded coding assignments and assisted students with OPL/CPLEX and Python Docplex implementations

TECHNICAL SKILLS

Programming: Python, R, SQL
Machine Learning & Statistical Libraries: statsmodels, scikit-learn, scikit-survival
Optimization & Simulation: Gurobi, Docplex, OPL/CPLEX, Simio (OptQuest)
Data Analysis & Visualization: NumPy, pandas, seaborn, matplotlib, Excel (VBA), Tableau
Platforms & Developer Tools: Google Cloud Platform, BigQuery, Git